

M02 - MLOps Reference Architecture



Topic

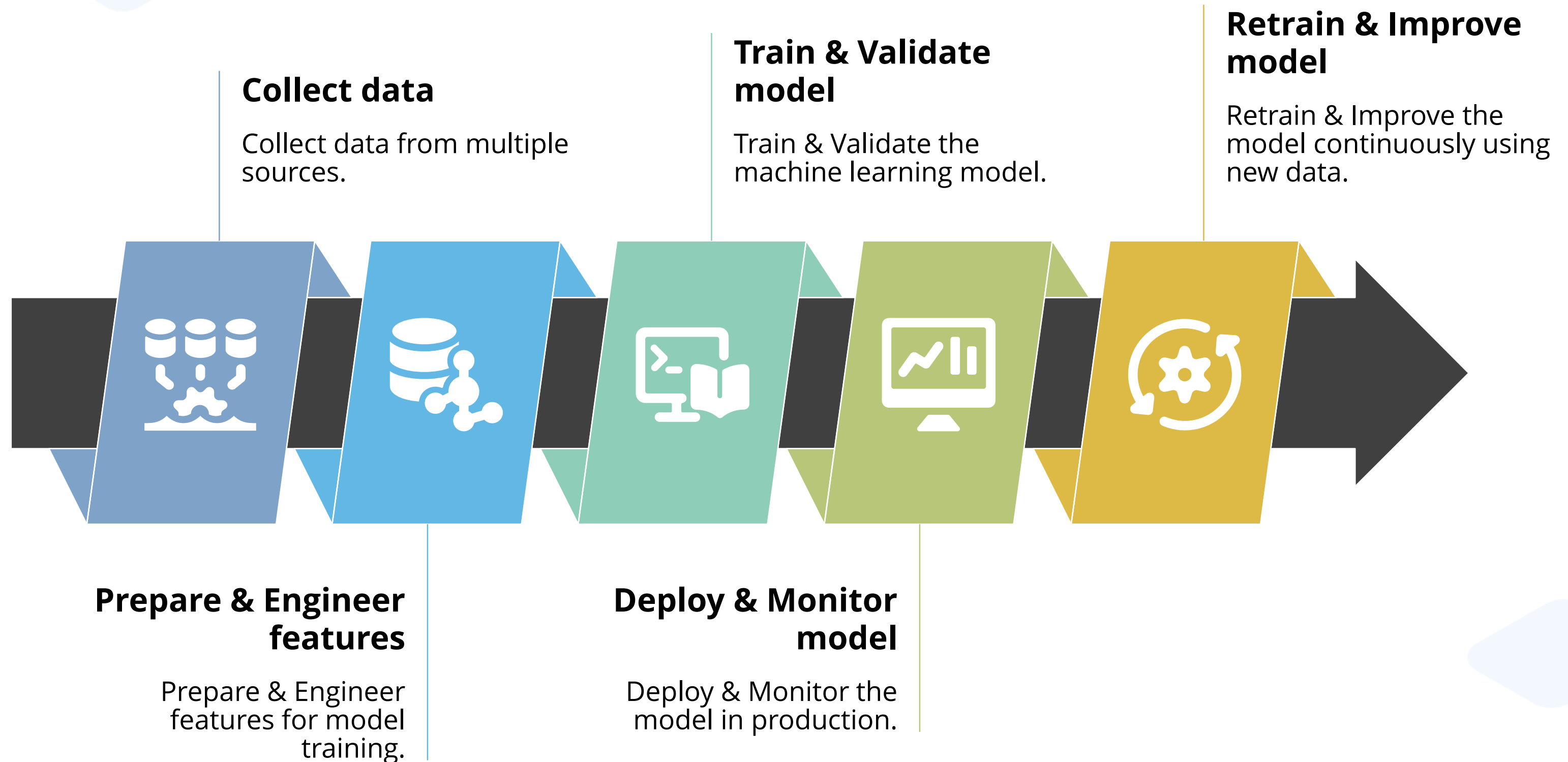
- **Understand the End-to-End MLOps Platform.**
- **Learn the purpose of the Data Layer.**
- **Explore the Feature Layer and feature management.**
- **Understand the Model Layer for training and versioning.**
- **Learn about the Serving Layer for model deployment.**
- **Compare Batch vs Real-Time Inference.**
- **Explore the Training Workflow.**
- **Understand the Deployment Workflow.**
- **Review Infrastructure Considerations for scalable MLOps.**
- **Explore Cloud Reference Architectures (Azure, AWS, GCP).**
- **Build a complete understanding of how all MLOps components work together.**





Understand the End-to-End MLOps Platform.

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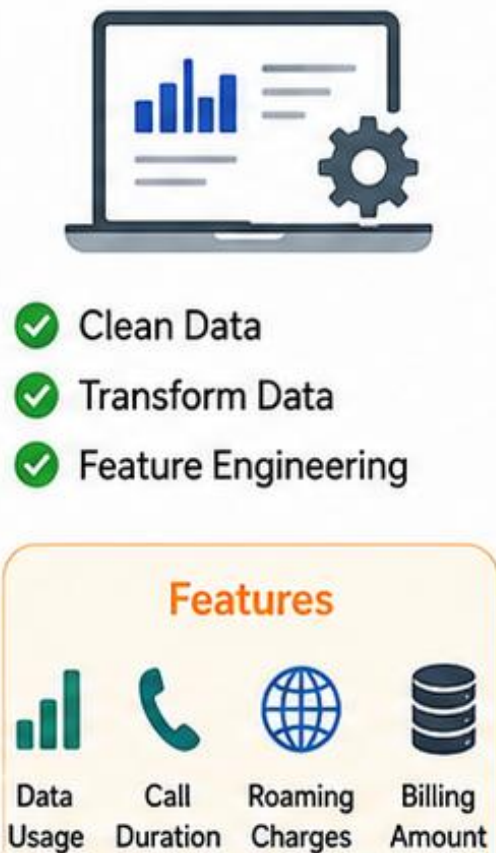
End-to-End MLOps Platform

Vodafone Billing Assurance

1 Collect Data



2 Prepare Data



3 Train & Validate



Train Model



Validate Model

4 Deploy



Model in Production



Used: 10 GB

Billed: 8 GB



Anomaly Detected

5 Monitor



Track Performance

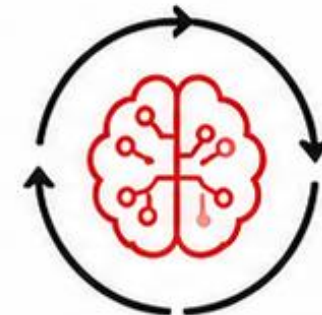


Detect Drift



Alerts

6 Retrain & Improve



Retrain with New Data



Redeploy Updated Model

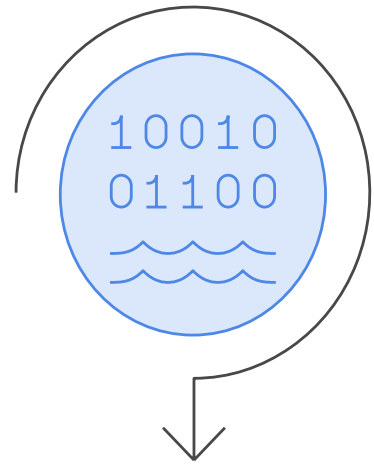
No



Performance Degraded?

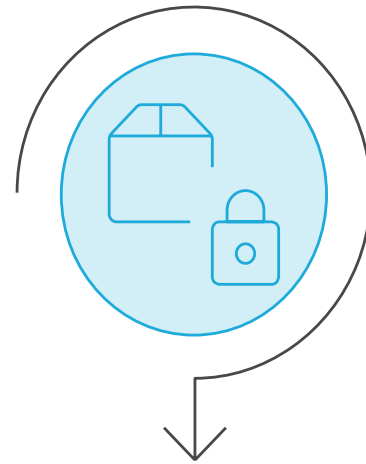
Yes

Purpose of the Data Layer



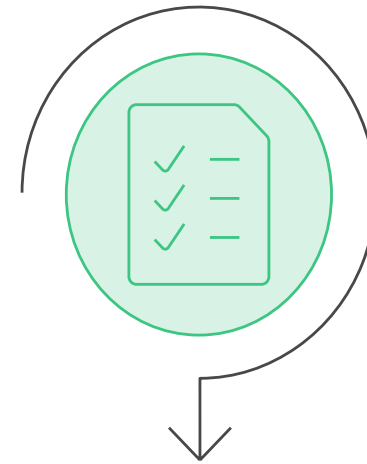
Collects data

Collects data from multiple business and operational systems.



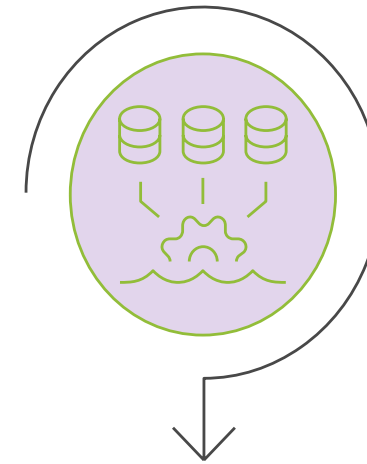
Stores data

Stores raw and processed data securely for analysis.



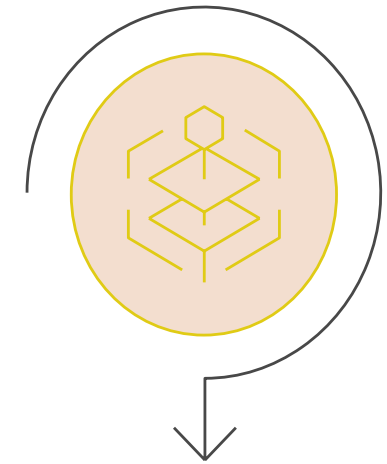
Ensures quality

Ensures data quality, consistency, and availability.



Supplies data

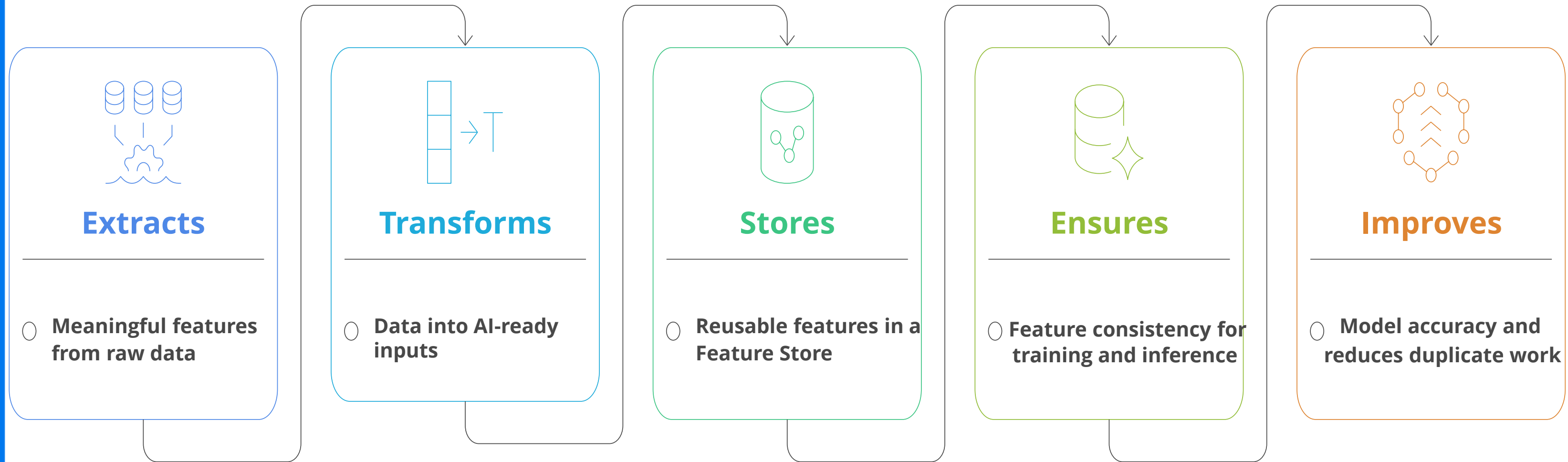
Supplies reliable data for feature engineering and model training.



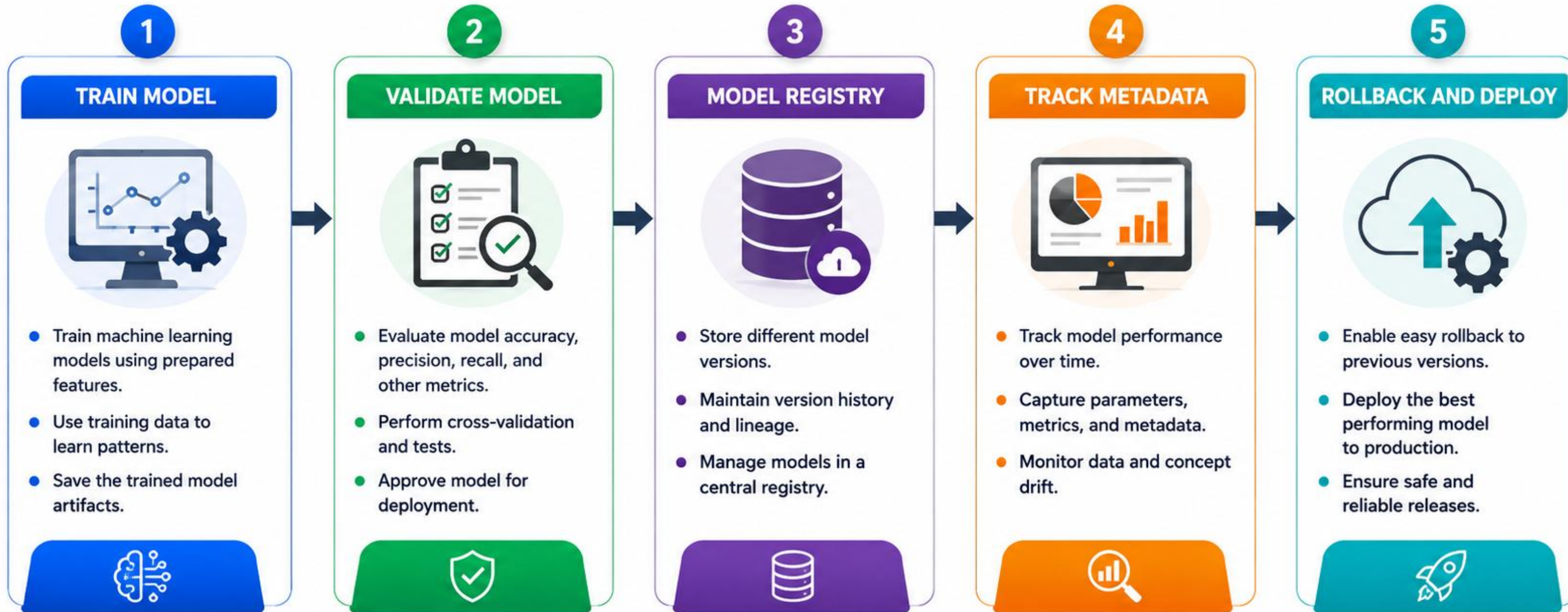
Acts as foundation

Acts as the foundation for the entire MLOps pipeline.

Feature Layer and Feature Management



Understand the Model Layer for training and versioning.



Serving Layer for Model Deployment



Deploys
trained models to
production.



Serves
real-time or batch
predictions.



Scales
automatically to handle
user requests.



Ensures
high availability,
security & low latency.



Supports
model updates and
rollback easily.

1. MODEL REGISTRY



Trained and versioned
models stored in the
Model Registry.

Example: Model v3.0
(Billing Anomaly Model)

2. DEPLOYMENT



Model is packaged and
deployed to the serving
infrastructure.

Docker Container /
Kubernetes / Cloud

3. MODEL SERVING

API ENDPOINT

REST/gRPC
Endpoint



PREDICTION

Returns
prediction
response



High Availability | Auto Scaling | Load Balancing



4. CONSUMERS



Web / Mobile Apps



Business Systems



Batch Processing



Streaming Platforms



EXAMPLE USE CASE

Billing Assurance –
Real-time Anomaly Detection



1. Live Data Arrives
Customer uses
10 GB data.



2. Request to
Serving Layer
Usage data is sent
to the model API.



3. Model Prediction
Model detects only
8 GB is billed.
Anomaly detected!



4. Response
Alert is returned
in milliseconds.



5. Action
Alert sent to Ops team
for investigation.

Batch vs Real-Time Inference

Batch Inference



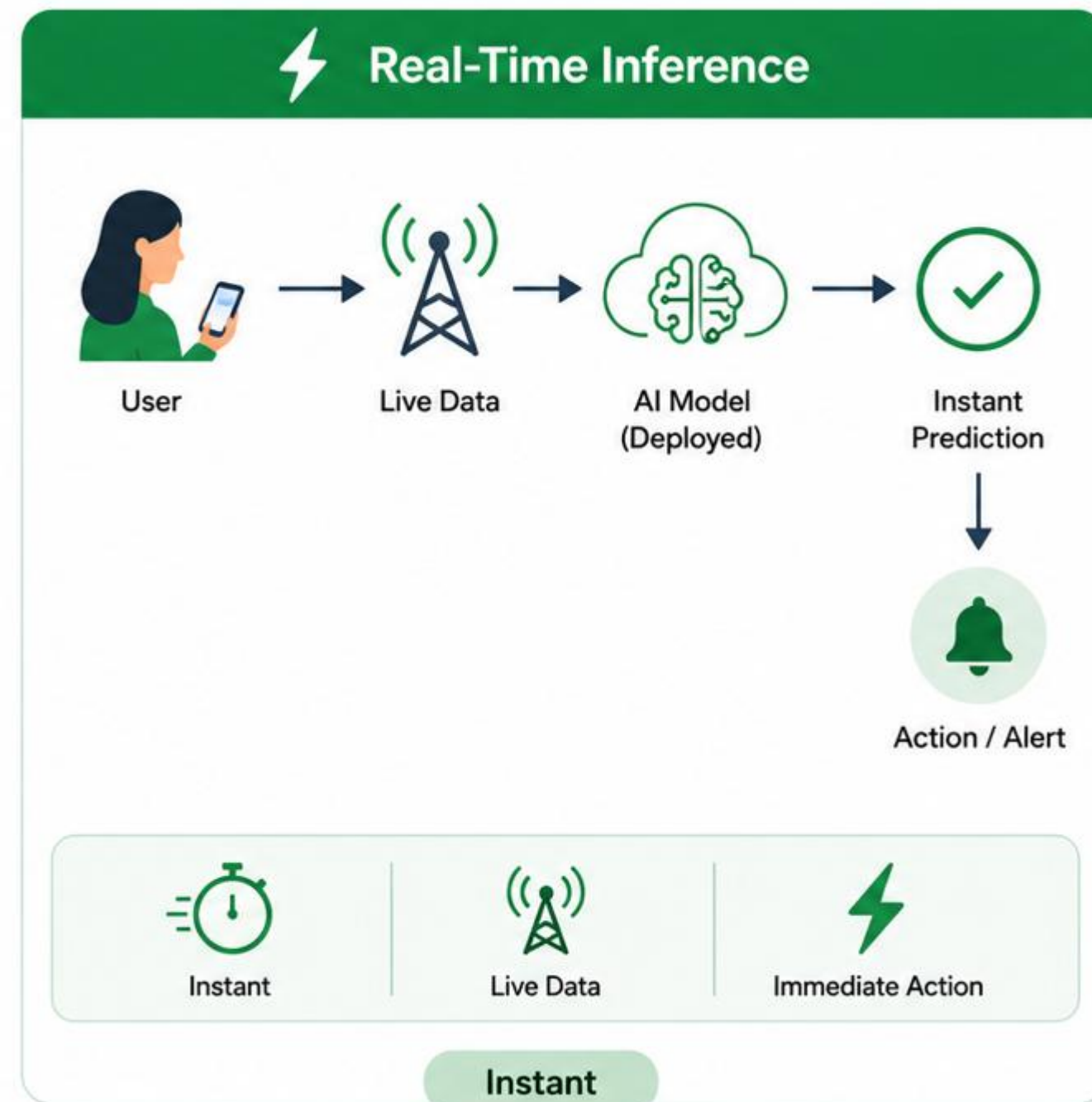
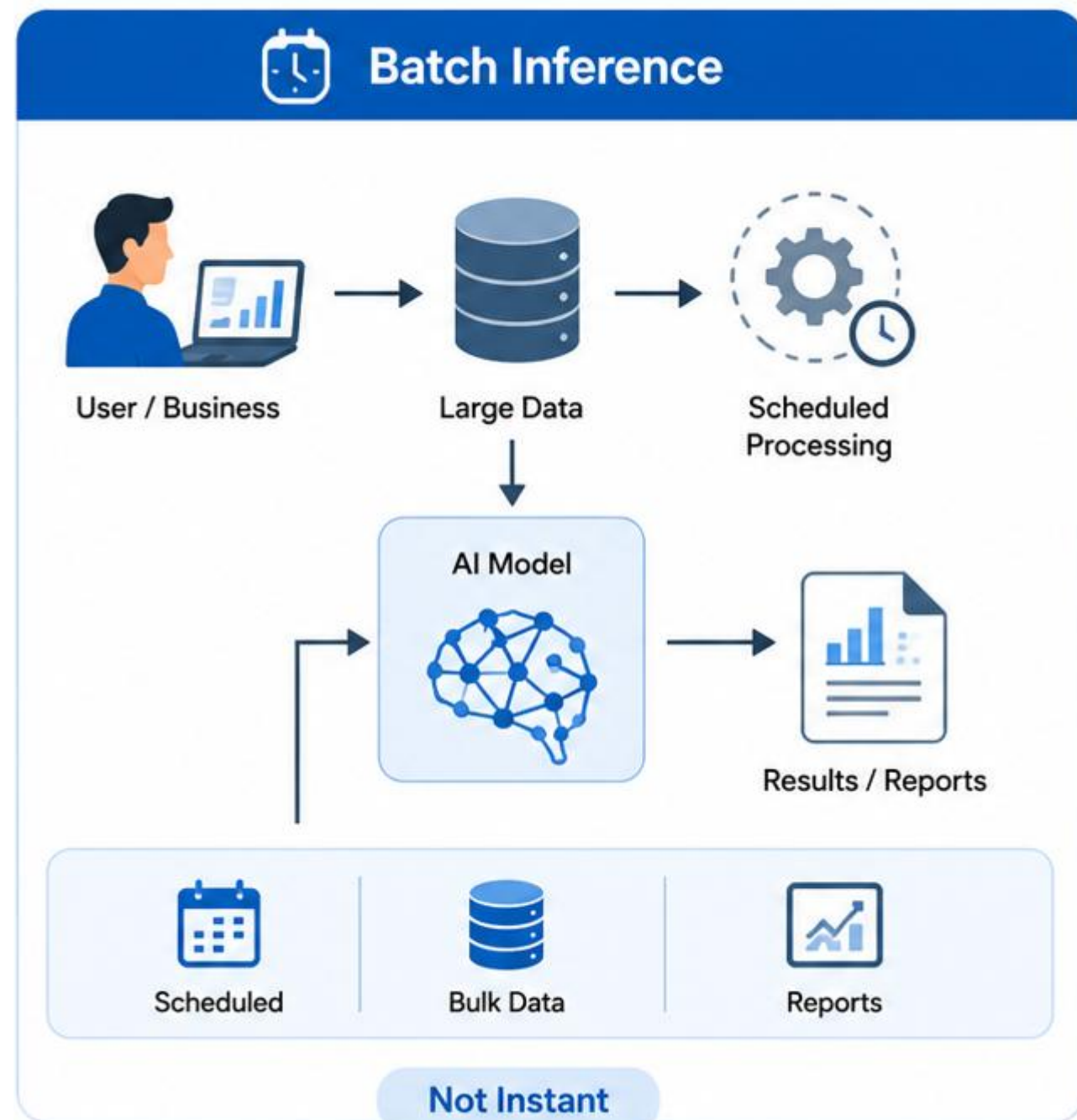
- Processes large amounts of data at scheduled intervals
- Ideal for reports, analytics, and periodic processing

Real-Time Inference

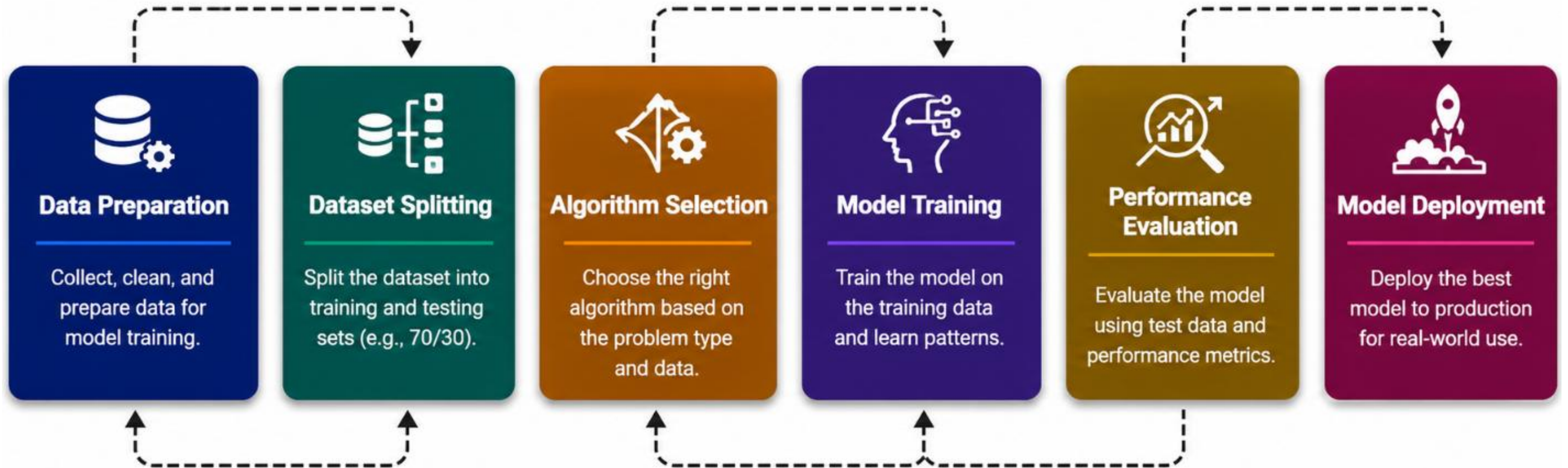


- Provides predictions instantly for live requests
- Used for fraud detection, recommendations, and live decisions

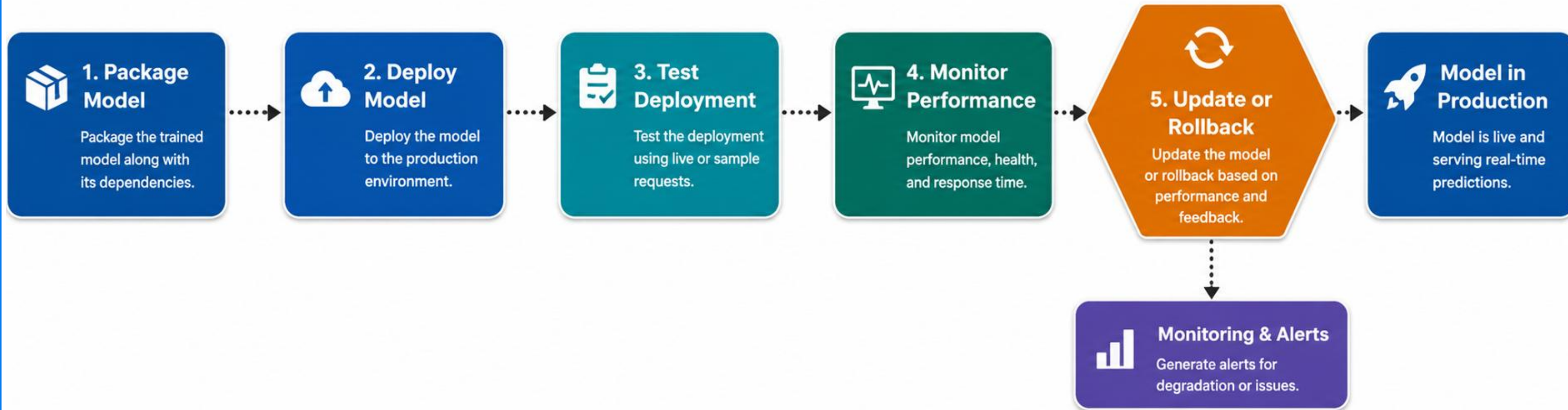
Batch vs Real-Time Inference- Vodafone Example



Training Workflow



Understand the Deployment Workflow.



Infrastructure Considerations for Scalable MLOps

Scalable Compute

Use cloud or Kubernetes to handle growing workloads.

Reliable Storage

Store datasets, models, and logs securely.

High Availability

Ensure AI services are always accessible.

Monitoring & Security

Track performance and protect data and models.

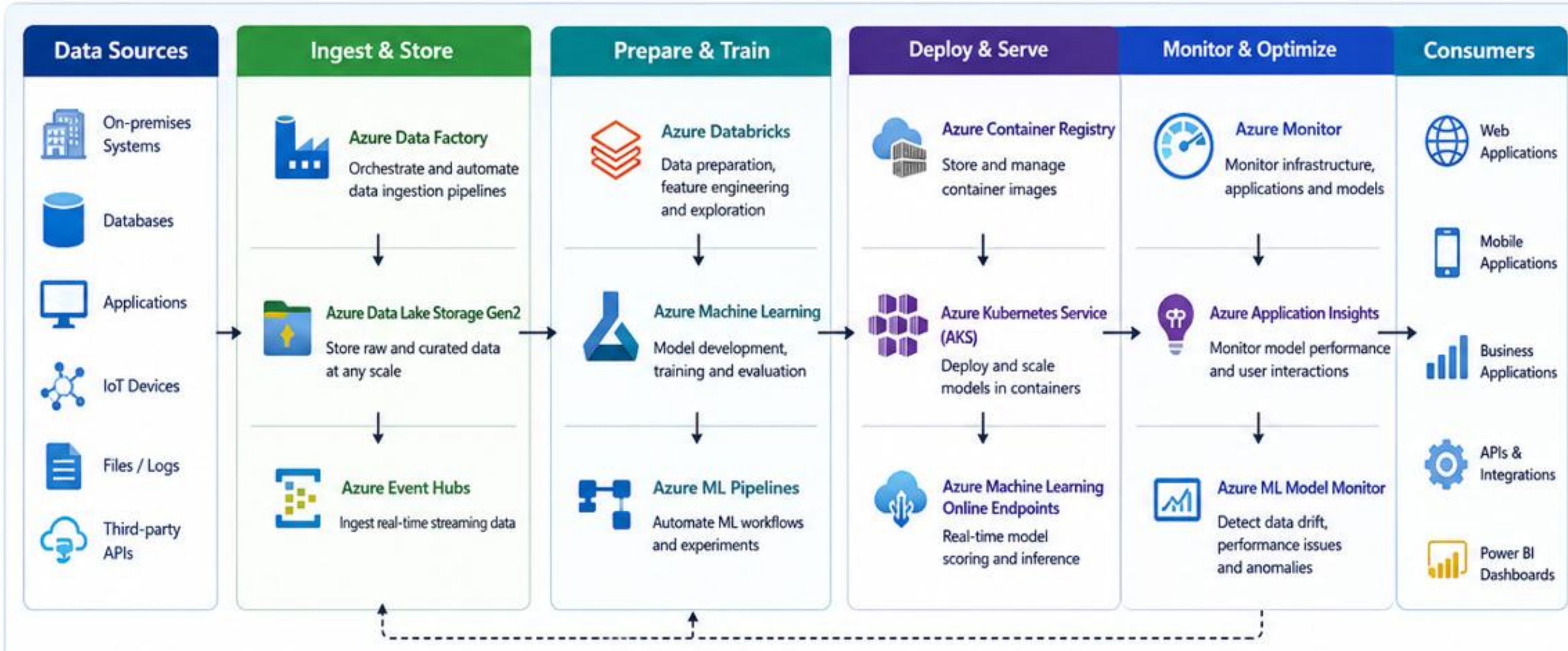
Auto Scaling

Automatically increase or decrease resources based on demand.

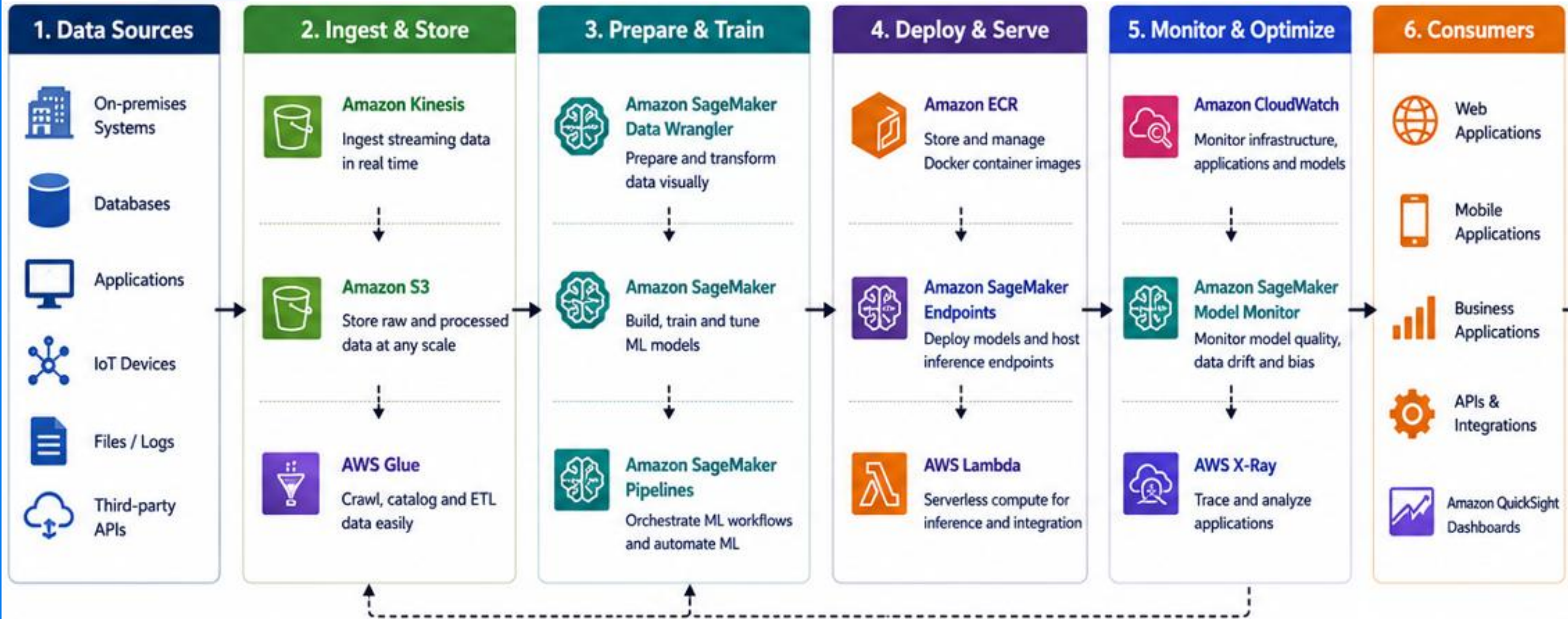


Explore Cloud Reference Architectures

Explore Cloud Reference Architectures -Azure



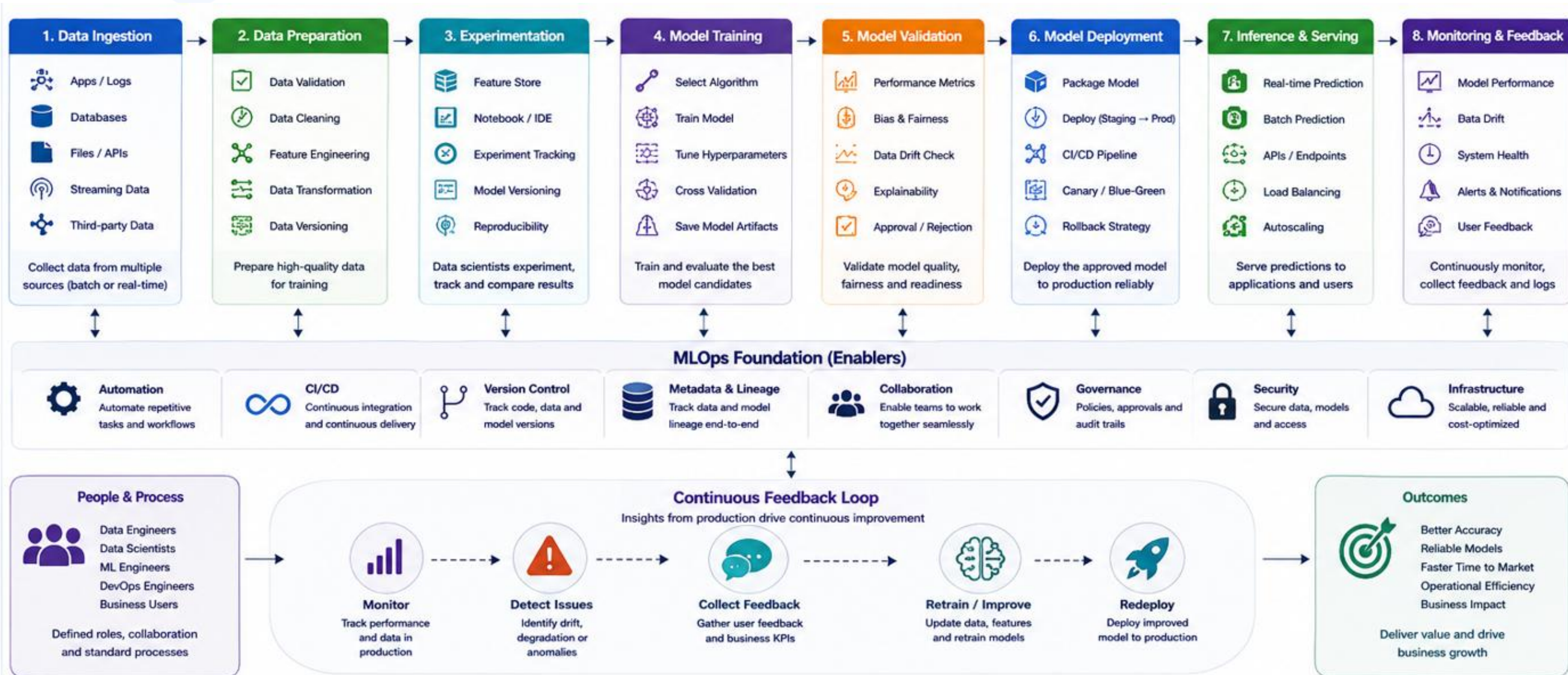
Explore Cloud Reference Architectures -AWS

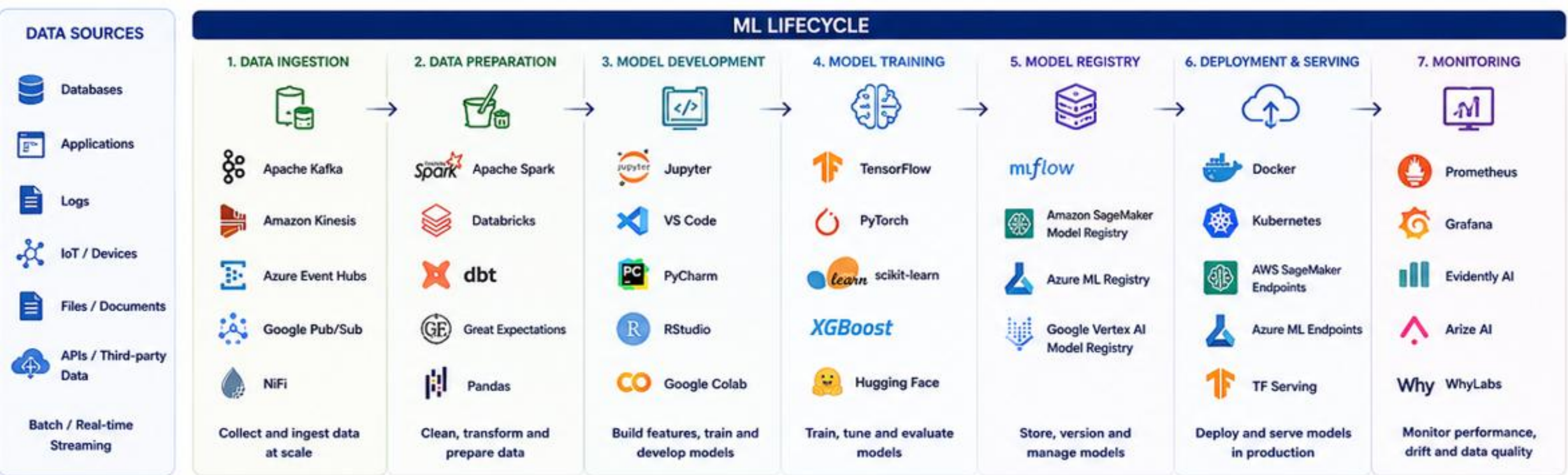




**Build a complete understanding of how all MLOps
components work together**

MLOps - End to End





QNA